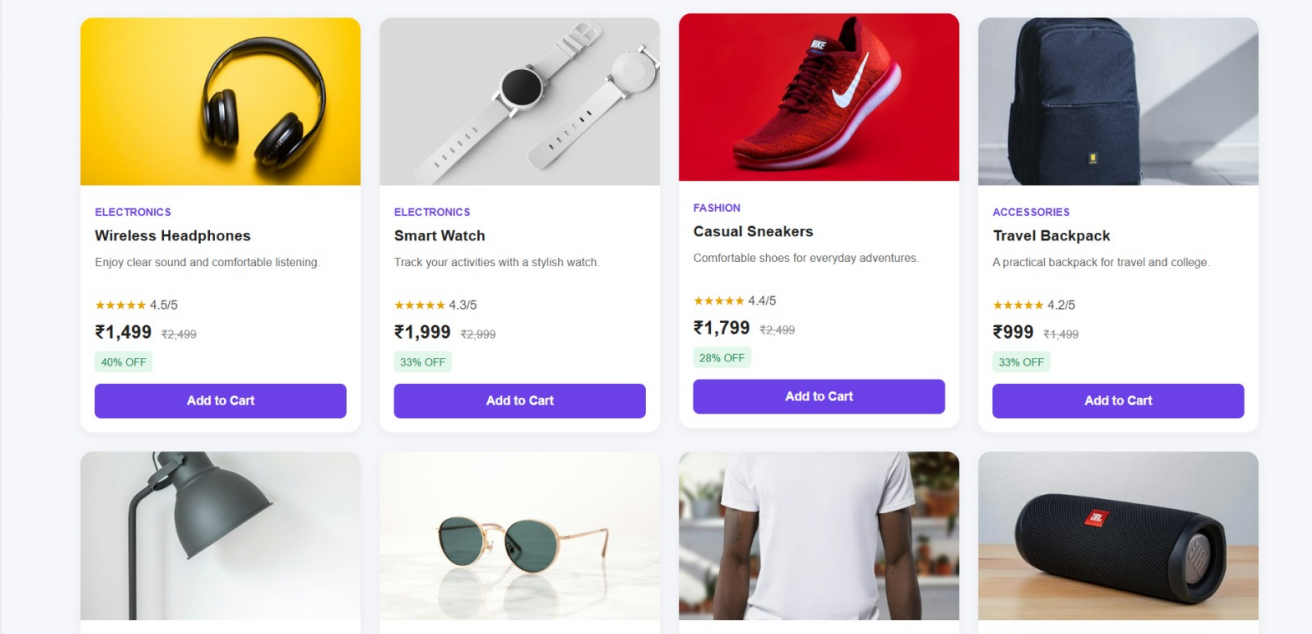
SHOPSMART

Online Shopping Website

Detailed Project Report, Website Output and Complete HTML, CSS & JavaScript Code



A responsive product-catalogue website inspired by the supplied reference screenshot. It presents products in clean cards with category labels, ratings, discounted prices and Add to Cart buttons.

Project type	Front-end web development demonstration
Technologies	HTML5, CSS3 and JavaScript
Purpose	Educational project / practical assignment

1. Introduction

ShopSmart is a responsive online shopping catalogue that demonstrates how a simple e-commerce front end can display a variety of products in a consistent and attractive layout. Each product card includes a photograph, category, product name, short description, rating, current price, original price, discount label and Add to Cart button.

The visual design follows the supplied reference: a light grey-lavender page, white rounded cards, product photographs, purple buttons, gold star ratings and green discount labels. The page is intended for learning and demonstration; it is not connected to a real inventory, payment service or delivery system.

2. Objectives

• Create an attractive and easy-to-navigate shopping catalogue.

• Present product details and pricing in a consistent card layout.

• Use category filters and a search field to help users find products.

• Make the layout adapt to desktop, tablet and mobile screen sizes.

• Add a simple interactive cart counter using JavaScript.

• Provide complete source code that can be saved and opened in a browser.

3. Main Website Features

Header and navigation: ShopSmart branding, navigation links and a cart counter sit in a clean top bar.

Product catalogue: Eight example items are arranged in a four-column grid on large screens and fewer columns on smaller screens.

Product cards: Each card displays an image, category, name, description, sample rating, sale price and crossed-out original price.

Discount labels: Green badges make example discounts easy to notice.

Search: Typing in the search box filters products by their product names.

Category filters: Buttons allow users to show all products or select Electronics, Fashion, Accessories or Home.

Add to Cart: Clicking a product button increases the cart count and displays a brief confirmation message.

Responsive design: CSS media queries change the number of columns and spacing for tablets and phones.

4. Product Catalogue

Product	Category	Sale price	Sample rating
Wireless Headphones	Electronics	■1,499	4.5/5
Smart Watch	Electronics	■1,999	4.3/5
Casual Sneakers	Fashion	■1,799	4.4/5
Travel Backpack	Accessories	■999	4.2/5
Desk Lamp	Home	■799	4.1/5
Round Sunglasses	Accessories	■699	4.0/5

Product	Category	Sale price	Sample rating
Basic Cotton T-Shirt	Fashion	■499	4.3/5
Bluetooth Speaker	Electronics	■1,299	4.6/5

Note: product names, prices, ratings and discounts are sample values for the assignment. They are not live offers or verified customer reviews.

5. Technologies Used

Technology	Purpose in this project
HTML5	Defines the page structure, navigation, product cards, search controls and footer.
CSS3	Controls colors, typography, spacing, grid layout, card appearance, hover states and responsive breakpoints.
JavaScript	Implements product search, category filtering, cart count updates and the confirmation message.

6. Design and Responsive Behaviour

Color palette: purple (#6d3ee8) for the primary buttons and highlights; pale lavender-grey (#f5f6fb) for the page; white for cards; gold (#e5a600) for star ratings; and pale green for discount badges.

CSS Grid: the product area uses repeat(4, minmax(0, 1fr)) on desktop. At widths below 1050px it uses three columns, below 760px it uses two, and below 460px it becomes a single column.

Image handling: product images fill the width of each card. object-fit: cover keeps the image area tidy while cropping edges when necessary.

Visual feedback: cards lift slightly on hover, buttons darken when hovered, and the cart confirmation appears temporarily after a click.

7. Development Steps

1. Plan the page structure and choose product categories.

2. Write semantic HTML for the header, toolbar, catalogue and footer.

3. Create CSS variables for reusable colors and consistent styles.

4. Build reusable product cards and arrange them with CSS Grid.

5. Add JavaScript event listeners for searching, filtering and cart feedback.

6. Test the layout at desktop, tablet and mobile widths, and check that each control responds.

8. Website Output

The image below is the reference screenshot supplied with the request. The implementation source code in the appendix recreates its main visual pattern: product cards, product photos, category names, ratings, sale and original prices, discount labels and purple Add to Cart buttons.

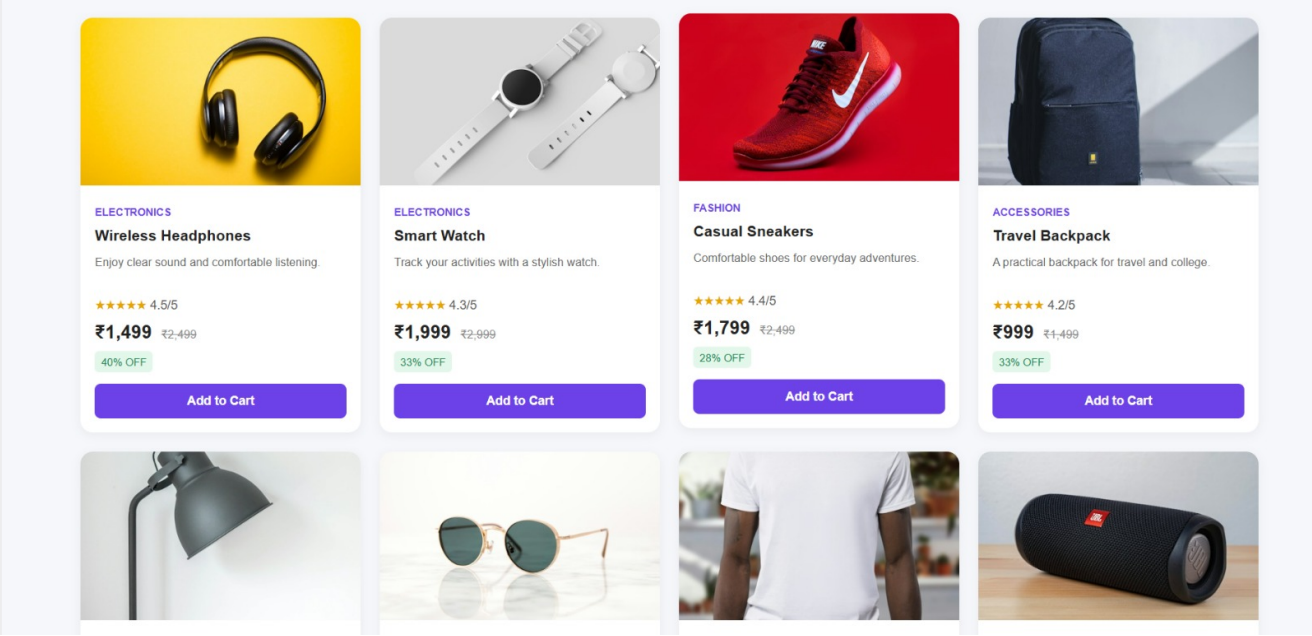


Figure 1. Supplied reference image for the shopping product grid.

9. How to Run the Website

1. Open a text editor such as Notepad or Visual Studio Code.

2. Copy the complete code from the appendix into a new file.

3. Save the file as index.html (choose All files if using Notepad).

4. Open index.html in Chrome, Edge, Firefox or another modern browser.

5. Use the search box and category buttons, then click Add to Cart to test the interactions.

An internet connection is needed for the external product photographs. If images do not load, check the connection or replace the image URLs with local image files.

10. Advantages, Limitations and Future Improvements

Advantages

• Clear product presentation and consistent card design.

• Search and category filters improve browsing.

• Responsive grid adapts to common screen sizes.

• Simple code structure is suitable for beginner-level study.

Limitations

• The cart is a demonstration counter and does not store item details or calculate totals.

• There is no login, checkout, payment processing, inventory or order tracking.

• Product information is hard-coded and sample-only.

• Images are loaded from external URLs and may require internet access.

Future improvements

A future version could include a cart summary with quantities and totals, product detail pages, sorting by price, user accounts, a checkout form, secure payment integration, order tracking, and a database or API for product data.

11. Conclusion

ShopSmart demonstrates the main building blocks of a simple online shopping interface. HTML structures the content, CSS creates the visual design and responsive layout, and JavaScript adds search, filtering and basic cart feedback. The project is a useful starting point for learning front-end web development, while a real e-commerce service would require secure back-end features.

12. Complete Source Code

The following appendix contains the full single-file implementation. Save it as index.html. HTML, CSS and JavaScript are included together so the example can be run without setting up a separate project folder.

Source code — index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>ShopSmart | Online Shopping</title>
  <style>
    :root {
      --purple: #6d3ee8;
      --purple-dark: #5127c9;
      --page: #f5f6fb;
      --text: #24232b;
      --muted: #777782;
      --gold: #e5a600;
      --green-bg: #e2f6e9;
      --green-text: #26784a;
      --white: #ffffff;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    html { scroll-behavior: smooth; }
    body {
      font-family: Arial, Helvetica, sans-serif;
      background: var(--page);
      color: var(--text);
      line-height: 1.5;
    }
    .topbar {
      background: #fff;
      padding: 18px 6%;
      display: flex;
      align-items: center;
      justify-content: space-between;
      gap: 18px;
      flex-wrap: wrap;
      box-shadow: 0 2px 10px #1717270b;
    }
    .brand { color: var(--purple); font-size: 25px; font-weight: 800; }
    .nav { display: flex; align-items: center; gap: 20px; flex-wrap: wrap; }
    .nav a { color: #454451; text-decoration: none; font-weight: 700; font-size: 14px; }
    .nav a:hover { color: var(--purple); }
    .cart-pill {
      background: var(--purple); color: #fff; border: 0; border-radius: 22px;
      padding: 10px 17px; font-weight: 700; cursor: pointer;
    }
    .intro { padding: 28px 6% 12px; }
    .intro h1 { font-size: clamp(26px, 3vw, 36px); margin-bottom: 6px; }
    .intro p { color: var(--muted); }
    .toolbar {
      padding: 12px 6% 22px; display: flex; gap: 12px; flex-wrap: wrap;
      align-items: center; justify-content: space-between;
    }
    .search {
      flex: 1; min-width: 210px; max-width: 430px; border: 1px solid #e0e0e9;
      background: #fff; border-radius: 10px; padding: 12px 14px; font-size: 14px;
    }
    .filters { display: flex; gap: 8px; flex-wrap: wrap; }
    .filter {
      border: 1px solid #dedee8; background: #fff; padding: 9px 13px;
      border-radius: 20px; cursor: pointer; color: #555461; font-weight: 700;
    }
    .filter.active, .filter:hover { background: var(--purple); color: #fff; border-color:
    var(--purple); }
    .product-grid {
      padding: 0 6% 45px; display: grid;
      grid-template-columns: repeat(4, minmax(0, 1fr)); gap: 22px;
    }
```

12. Complete Source Code (continued — lines 67 onward)

```
    .product-card {
      background: #fff; border-radius: 16px; overflow: hidden;
      box-shadow: 0 5px 18px #25243b0b; display: flex; flex-direction: column;
      min-width: 0; transition: transform .2s, box-shadow .2s;
    }
    .product-card:hover { transform: translateY(-4px); box-shadow: 0 10px 24px #25243b15; }
    .product-card img { width: 100%; height: 205px; object-fit: cover; display: block; }
    .product-info { padding: 17px 17px 18px; display: flex; flex-direction: column; flex: 1; }
    .category { color: var(--purple); font-size: 12px; font-weight: 800; letter-spacing: .3px;
    margin-bottom: 5px; }
    .product-info h2 { font-size: 18px; line-height: 1.3; margin-bottom: 7px; }
    .description { color: var(--muted); font-size: 13px; min-height: 39px; margin-bottom:
    12px; }
    .rating { font-size: 14px; color: #696873; margin-bottom: 8px; }
    .stars { color: var(--gold); letter-spacing: -1px; }
    .price-row { display: flex; align-items: baseline; gap: 9px; margin-bottom: 8px; }
    .price { font-size: 22px; font-weight: 800; }
    .old-price { color: #9999a1; font-size: 12px; text-decoration: line-through; }
    .discount {
      align-self: flex-start; background: var(--green-bg); color: var(--green-text);
      border-radius: 5px; padding: 3px 8px; font-size: 12px; margin-bottom: 13px;
    }
    .add-btn {
      margin-top: auto; width: 100%; border: 0; background: var(--purple); color: #fff;
      border-radius: 8px; padding: 11px 12px; font-size: 14px; font-weight: 800; cursor:
    pointer;
    }
    .add-btn:hover { background: var(--purple-dark); }
    .empty { grid-column: 1 / -1; text-align: center; color: var(--muted); padding: 40px; }
    .cart-message {
      position: fixed; right: 20px; bottom: 20px; z-index: 5; background: #29243a;
      color: #fff; padding: 13px 18px; border-radius: 10px; box-shadow: 0 5px 20px #0002;
      display: none;
    }
    footer { text-align: center; padding: 24px; background: #fff; color: var(--muted);
    font-size: 13px; }
    @media (max-width: 1050px) { .product-grid { grid-template-columns: repeat(3, minmax(0,
    1fr)); } }
    @media (max-width: 760px) {
      .product-grid { grid-template-columns: repeat(2, minmax(0, 1fr)); gap: 14px; }
      .product-card img { height: 175px; }
      .topbar { justify-content: center; }
      .nav { justify-content: center; }
      .intro, .toolbar { padding-left: 5%; padding-right: 5%; }
      .product-grid { padding-left: 5%; padding-right: 5%; }
    }
    @media (max-width: 460px) {
      .product-grid { grid-template-columns: 1fr; }
      .product-card img { height: 220px; }
      .product-info h2 { font-size: 19px; }
      .description { min-height: 0; }
    }
  </style>
</head>
<body>
  <header class="topbar">
    <div class="brand">ShopSmart</div>
    <nav class="nav" aria-label="Main navigation">
      <a href="#home">Home</a>
      <a href="#products">Products</a>
      <a href="#electronics">Electronics</a>
      <a href="#fashion">Fashion</a>
      <a href="#accessories">Accessories</a>
    </nav>
    <button class="cart-pill" type="button" aria-label="Shopping cart">
      Cart (<span id="cart-count">0</span>)
```

12. Complete Source Code (continued — lines 133 onward)

```
    </button>
  </header>

  <main id="home">
    <section class="intro">
      <h1>Find your everyday favourites</h1>
      <p>Great products, clear prices and special deals—all in one place.</p>
    </section>

    <section class="toolbar" aria-label="Product search and filters">
      <input id="search" class="search" type="search"
             placeholder="Search products..." aria-label="Search products">
      <div class="filters">
        <button class="filter active" data-filter="all" type="button">All</button>
        <button class="filter" data-filter="electronics" type="button">Electronics</button>
        <button class="filter" data-filter="fashion" type="button">Fashion</button>
        <button class="filter" data-filter="accessories" type="button">Accessories</button>
        <button class="filter" data-filter="home" type="button">Home</button>
      </div>
    </section>

    <section class="product-grid" id="products" aria-label="Product catalogue">
      <article class="product-card" data-category="electronics" data-name="wireless
    headphones">
        <img src="https://images.unsplash.com/photo-1505740420928-5e560c06d30e?auto=format&fit
    =crop&w=800&q=80"
             alt="Wireless headphones">
        <div class="product-info">
          <p class="category">ELECTRONICS</p><h2>Wireless Headphones</h2>
          <p class="description">Enjoy clear sound and comfortable listening.</p>
          <p class="rating"><span class="stars">★★★★★</span> 4.5/5</p>
          <div class="price-row"><span class="price">■1,499</span><span
    class="old-price">■2,499</span></div>
          <span class="discount">40% OFF</span>
          <button class="add-btn" data-product="Wireless Headphones" data-price="1499">Add to
    Cart</button>
        </div>
      </article>

      <article class="product-card" data-category="electronics" data-name="smart watch">
        <img src="https://images.unsplash.com/photo-1523275335684-37898b6baf30?auto=format&fit
    =crop&w=800&q=80" alt="Smart watch">
        <div class="product-info">
          <p class="category">ELECTRONICS</p><h2>Smart Watch</h2>
          <p class="description">Track your activities with a stylish watch.</p>
          <p class="rating"><span class="stars">★★★★★</span> 4.3/5</p>
          <div class="price-row"><span class="price">■1,999</span><span
    class="old-price">■2,999</span></div>
          <span class="discount">33% OFF</span>
          <button class="add-btn" data-product="Smart Watch" data-price="1999">Add to
    Cart</button>
        </div>
      </article>

      <article class="product-card" data-category="fashion" data-name="casual sneakers">
        <img src="https://images.unsplash.com/photo-1542291026-7eec264c27ff?auto=format&fit=cr
    op&w=800&q=80" alt="Red casual sneaker">
        <div class="product-info">
          <p class="category">FASHION</p><h2>Casual Sneakers</h2>
          <p class="description">Comfortable shoes for everyday adventures.</p>
          <p class="rating"><span class="stars">★★★★★</span> 4.4/5</p>
          <div class="price-row"><span class="price">■1,799</span><span
    class="old-price">■2,499</span></div>
          <span class="discount">28% OFF</span>
          <button class="add-btn" data-product="Casual Sneakers" data-price="1799">Add to
    Cart</button>
```

12. Complete Source Code (continued — lines 199 onward)

```
      </div>
    </article>

    <article class="product-card" data-category="accessories" data-name="travel backpack">
      <img src="https://images.unsplash.com/photo-1553062407-98eeb64c6a62?auto=format&fit=cr
op&w=800&q=80" alt="Travel backpack">
      <div class="product-info">
        <p class="category">ACCESSORIES</p><h2>Travel Backpack</h2>
        <p class="description">A practical backpack for travel and college.</p>
        <p class="rating"><span class="stars">★★★★★</span> 4.2/5</p>
        <div class="price-row"><span class="price">■999</span><span
class="old-price">■1,499</span></div>
        <span class="discount">33% OFF</span>
        <button class="add-btn" data-product="Travel Backpack" data-price="999">Add to
Cart</button>
      </div>
    </article>

    <article class="product-card" data-category="home" data-name="desk lamp">
      <img src="https://images.unsplash.com/photo-1507473885765-e6ed057f782c?auto=format&fit
=crop&w=800&q=80" alt="Modern desk lamp">
      <div class="product-info">
        <p class="category">HOME</p><h2>Desk Lamp</h2>
        <p class="description">A simple lamp for reading and study spaces.</p>
        <p class="rating"><span class="stars">★★★★★</span> 4.1/5</p>
        <div class="price-row"><span class="price">■799</span><span
class="old-price">■1,199</span></div>
        <span class="discount">33% OFF</span>
        <button class="add-btn" data-product="Desk Lamp" data-price="799">Add to
Cart</button>
      </div>
    </article>

    <article class="product-card" data-category="accessories" data-name="round sunglasses">
      <img src="https://images.unsplash.com/photo-1511499767150-a48a237f0083?auto=format&fit
=crop&w=800&q=80" alt="Round sunglasses">
      <div class="product-info">
        <p class="category">ACCESSORIES</p><h2>Round Sunglasses</h2>
        <p class="description">Classic frames for sunny days outdoors.</p>
        <p class="rating"><span class="stars">★★★★★</span> 4.0/5</p>
        <div class="price-row"><span class="price">■699</span><span
class="old-price">■999</span></div>
        <span class="discount">30% OFF</span>
        <button class="add-btn" data-product="Round Sunglasses" data-price="699">Add to
Cart</button>
      </div>
    </article>

    <article class="product-card" data-category="fashion" data-name="basic cotton t shirt">
      <img src="https://images.unsplash.com/photo-1521572163474-6864f9cf17ab?auto=format&fit
=crop&w=800&q=80" alt="White cotton T-shirt">
      <div class="product-info">
        <p class="category">FASHION</p><h2>Basic Cotton T-Shirt</h2>
        <p class="description">A soft, versatile T-shirt for daily wear.</p>
        <p class="rating"><span class="stars">★★★★★</span> 4.3/5</p>
        <div class="price-row"><span class="price">■499</span><span
class="old-price">■799</span></div>
        <span class="discount">38% OFF</span>
        <button class="add-btn" data-product="Basic Cotton T-Shirt" data-price="499">Add to
Cart</button>
      </div>
    </article>

    <article class="product-card" data-category="electronics" data-name="portable bluetooth
speaker">
      <img src="https://images.unsplash.com/photo-1608043152269-423dbba4e7e1?auto=format&fit
```

12. Complete Source Code (continued — lines 265 onward)

```
  =crop&w=800&q=80" alt="Portable Bluetooth speaker">
      <div class="product-info">
        <p class="category">ELECTRONICS</p><h2>Bluetooth Speaker</h2>
        <p class="description">Portable sound for music at home or outdoors.</p>
        <p class="rating"><span class="stars">★★★★★</span> 4.6/5</p>
        <div class="price-row"><span class="price">■1,299</span><span
  class="old-price">■1,999</span></div>
        <span class="discount">35% OFF</span>
        <button class="add-btn" data-product="Bluetooth Speaker" data-price="1299">Add to
  Cart</button>
      </div>
    </article>
  </section>
  <div class="cart-message" id="cart-message" role="status" aria-live="polite"></div>
</main>

<footer>
  <p>© 2026 ShopSmart. Demo shopping website for educational use.</p>
  <p>Product prices, ratings and discounts are sample information.</p>
</footer>

<script>
  const cards = [...document.querySelectorAll(".product-card")];
  const searchInput = document.getElementById("search");
  const filterButtons = [...document.querySelectorAll(".filter")];
  const cartCount = document.getElementById("cart-count");
  const cartMessage = document.getElementById("cart-message");
  let activeCategory = "all";
  let count = 0;

  function filterProducts() {
    const term = searchInput.value.trim().toLowerCase();
    let visible = 0;
    cards.forEach(card => {
      const matchesCategory = activeCategory === "all" ||
        card.dataset.category === activeCategory;
      const matchesSearch = card.dataset.name.includes(term);
      card.hidden = !(matchesCategory && matchesSearch);
      if (!card.hidden) visible++;
    });
    const oldEmpty = document.querySelector(".empty");
    if (oldEmpty) oldEmpty.remove();
    if (visible === 0) {
      const message = document.createElement("p");
      message.className = "empty";
      message.textContent = "No products found. Try another search or category.";
      document.getElementById("products").appendChild(message);
    }
  }

  filterButtons.forEach(button => {
    button.addEventListener("click", () => {
      activeCategory = button.dataset.filter;
      filterButtons.forEach(item => item.classList.toggle("active", item === button));
      filterProducts();
    });
  });
  searchInput.addEventListener("input", filterProducts);

  document.querySelectorAll(".add-btn").forEach(button => {
    button.addEventListener("click", () => {
      count++;
      cartCount.textContent = count;
      cartMessage.textContent = button.dataset.product + " added to cart!";
      cartMessage.style.display = "block";
      window.clearTimeout(window.shopMessageTimer);
```

12. Complete Source Code (continued — lines 331 onward)

```
        window.shopMessageTimer = window.setTimeout(() => {
          cartMessage.style.display = "none";
        }, 2200);
      });
    });
  </script>
</body>
</html>
```